

The 1984 Data Protection Act: The 1st data protection act to protect individuals data against:

- * The use of inaccurate personal information or information that is incomplete or irrelevant
 - * The use of personal information by unauthorised persons
 - * The use of personal information for purposes other than that for which it was collected
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Summary of The BCS Code of Conduct:

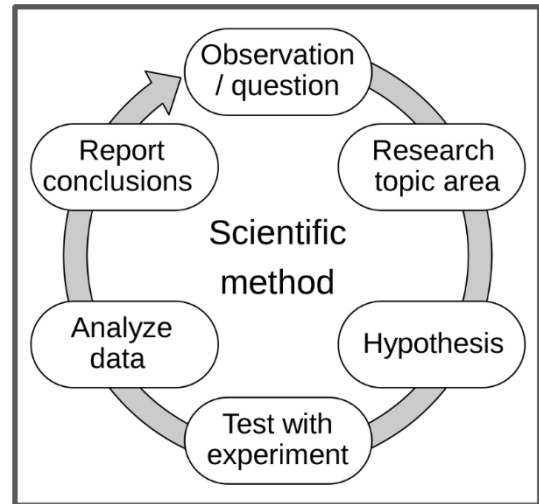
- The Code sets out the ethical and professional standards expected of all members of the British Computer Society (BCS).
 - It ensures members act responsibly, with integrity, and in the public interest.
 - It applies to all professional, academic, and voluntary activities undertaken by BCS members.
 - Breaches can result in disciplinary action or loss of membership.
1. Public Interest. Members must:
 - Respect and safeguard public health, safety, privacy, and wellbeing.
 - Avoid harm to others and the environment through their work.
 - Respect third-party rights, including intellectual property and data protection.
 - Promote equal access to the benefits of IT; avoid unfair discrimination.
 - Comply with relevant laws and regulations.
 2. Professional Competence and Integrity. Members must:
 - Work within their competence and only undertake tasks they are qualified to perform.
 - Maintain and update their professional knowledge and skills.
 - Be honest and accurate in representing their professional experience and qualifications.
 - Avoid conflicts of interest and disclose them when unavoidable.
 - Reject deceptive or unethical practices, such as plagiarism or data falsification.
 - Acknowledge and correct errors promptly when discovered.
 3. Duty to Relevant Authority. Members must:
 - Act in the best interests of their employer, client, or academic institution, consistent with the public good.
 - Protect confidential information and not disclose it without proper authorisation.
 - Provide accurate and objective advice, even if it is unwelcome.
 - Avoid misrepresentation of system performance, costs, or benefits.
 - Report unethical or unlawful practices where appropriate (whistleblowing responsibility).
 4. Duty to the Profession. Members must:
 - Uphold the reputation of the computing profession and the BCS.
 - Promote understanding of IT's impact on society and responsible technology use.
 - Support the professional development of colleagues and peers.
 - Behave with integrity and respect in professional interactions.
 - Avoid bringing the profession into disrepute through personal or professional misconduct.
 - Encourage ethical conduct among peers and future professionals.
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UK Data Protection Act (in brief):

1. Personal data shall be processed fairly and lawfully, and not without the subject's consent.
 2. Personal data shall only be processed for specified and lawful purposes (and not for any other).
 3. Personal data should be adequate, relevant and not excessive in relation to the purpose.
 4. Personal data must be accurate and up to date.
 5. Personal data should not be kept for longer than is necessary for purpose.
 6. Personal data must take into account the rights of subjects (see later...).
 7. Technical & organisational operations must ensure there is no unlawful processing or loss of personal data.
 8. Personal data shall not be transferred to a country outside the EU unless it has similar data protection laws.
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Scientific method Stages:

1. Define a question
2. Gather information and resources (observe)
3. Form an explanatory hypothesis
4. Test the hypothesis by performing an experiment and collecting data in a reproducible manner
5. Analyze the data
6. Interpret the data and draw conclusions that serve as a starting point for a new hypothesis
7. Publish results
8. Retest (frequently done by other scientists)

**Experimental factors checklist:**

- Research question/ hypothesis. – Independent/ dependent variables. – Control variables. -Random variables. – Confounding factors. – Participants. – Within/ between subject design.

A hypothesis: a specific and testable explanation of a phenomenon. A hypothesis makes a prediction and is an attempt to explain an observation that cannot be adequately described by existing accepted scientific theories.

Independent variables: What you deliberately change in the experiment - A variable that the experimenter controls

Dependent variable: the data you are collecting. A variable whose value(s) depend on the independent variable

Control Variables: factors that do not change to ensure a “fair” comparison

Random Variables: factors you knowingly allow to randomly change/vary

Confounding Factors: factors that change and which impact the results (usually only identified after you have collected the data)

Experimental Design:

- Experiments are used to gather empirical data/ measurements
- Experimental design vary – all serve the same purpose: gathering data against which we can test our hypotheses.

Within-subject design: All subjects are exposed to all conditions;

-i.e., they experience every level of the independent variable(s);

-Subjects are the things you are studying: people, images, .. etc

Also known as: repeated-measure design; because you repeat the measurements for every condition.

Between-subject design: subjects are placed into groups and exposed to just one condition;

-i.e., they experience one level of the independent variable(s);

Also known as between-groups or independent-measures design;

Because you make independent measurements for each condition.

Type of Experimental data: quantitative data vs qualitative data.

Types of data: continuous data, discrete data, nominal data, ordinal data, interval data, ratio data.